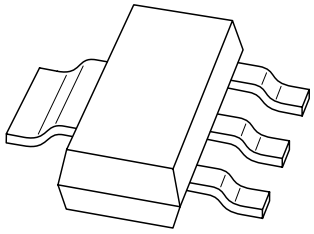
DATA SHEET



BSP60; BSP61; BSP62
PNP Darlington transistors

Product specification
Supersedes data of 1997 Apr 22

1999 Apr 29

Philips
Semiconductors

PHILIPS

PNP Darlington transistors BSP60; BSP61; BSP62

FEATURES

- High current (max. 0.5 A)
- Low voltage (max. 80 V)
- Integrated diode and resistor.

APPLICATIONS

- Industrial switching applications such as:
  - Print hammer
  - Solenoid
  - Relay and lamp drivers.

DESCRIPTION

PNP Darlington transistor in a SOT223 plastic package.
NPN complements: BSP50, BSP51 and BSP52.

PINNING

PIN	DESCRIPTION
1	base
2,4	collector
3	emitter

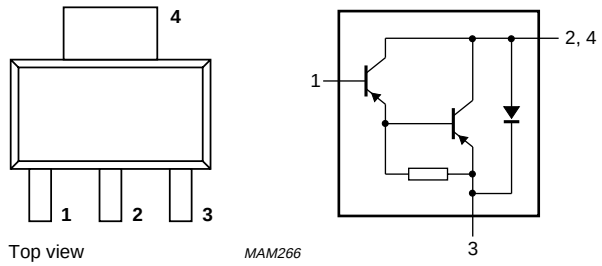


Fig.1 Simplified outline (SOT223) and symbol.

LIMITING VALUES

In accordance with the Absolute Maximum Rating System (IEC 134).

SYMBOL	PARAMETER	CONDITIONS	MIN.	MAX.	UNIT
V_{CBO}	collector-base voltage	open emitter			
	BSP60		–	–60	V
	BSP61		–	–80	V
	BSP62		–	–90	V
V_{CES}	collector-emitter voltage	V_{BE} = 0			
	BSP60		–	–45	V
	BSP61		–	–60	V
	BSP62		–	–80	V
V_{EBO}	emitter-base voltage	open collector	–	–5	V
I_C	collector current (DC)		–	–0.5	A
I_{CM}	peak collector current		–	–1.5	A
I_B	base current (DC)		–	–100	mA
P_{tot}	total power dissipation	$T_{amb} \leq 25$ °C; note 1	–	1.25	W
T_{stg}	storage temperature		–65	+150	°C
T_j	junction temperature		–	150	°C
T_{amb}	operating ambient temperature		–65	+150	°C

Note

1. Device mounted on a printed-circuit board, single sided copper, tinplated, mounting pad for collector 1 cm^2. For other mounting conditions, see *"Thermal considerations for the SOT223 in the General Part of associated Handbook".*

THERMAL CHARACTERISTICS

SYMBOL	PARAMETER	CONDITIONS	VALUE	UNIT
$R_{th\ j\text{-}a}$	thermal resistance from junction to ambient	note 1	98	K/W
$R_{th\ j\text{-}s}$	thermal resistance from junction to solder point		17	K/W

Note

1. Device mounted on a printed-circuit board, single sided copper, tinplated, mounting pad for collector 1 cm^2. For other mounting conditions, see *"Thermal considerations for the SOT223 in the General Part of associated Handbook".*

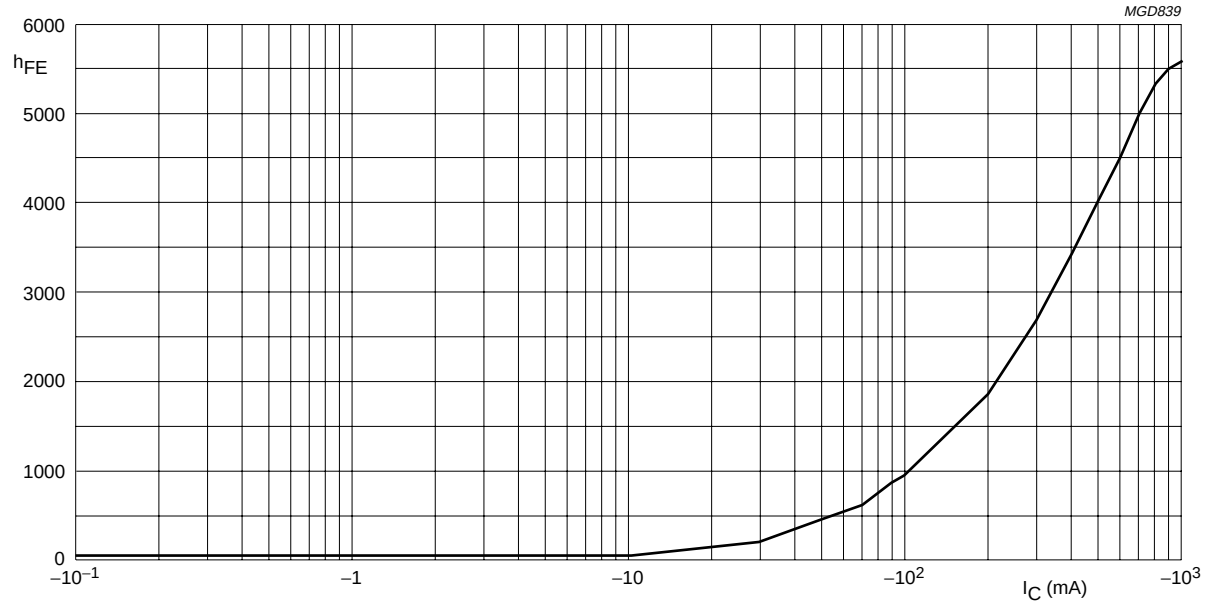
CHARACTERISTICS

T_j = 25 °C unless otherwise specified.

SYMBOL	PARAMETER	CONDITIONS	MIN.	TYP.	MAX.	UNIT
I_{CES}	collector cut-off current					
	BSP60	V_{BE} = 0; V_{CE} = –45 V	–	–	–50	nA
	BSP61	V_{BE} = 0; V_{CE} = –60 V	–	–	–50	nA
	BSP62	V_{BE} = 0; V_{CE} = –80 V	–	–	–50	nA
I_{EBO}	emitter cut-off current	I_C = 0; V_{EB} = –4 V	–	–	–50	nA
h_{FE}	DC current gain	V_{CE} = –10 V; note 1; see Fig.2				
		I_C = –150 mA	1000	–	–	
		I_C = –500 mA	2000	–	–	
V_{CEsat}	collector-emitter saturation voltage	I_C = –500 mA; I_B = –0.5 mA	–	–	–1.3	V
		I_C = –500 mA; I_B = –0.5 mA; T_j = 150 °C	–	–	–1.3	V
V_{BEsat}	base-emitter saturation voltage	I_C = –500 mA; I_B = –0.5 mA	–	–	–1.9	V
f_T	transition frequency	I_C = –500 mA; V_{CE} = –5 V; f = 100 MHz	–	200	–	MHz
Switching times (between 10% and 90% levels); see Fig.3						
t_{on}	turn-on time	I_{Con} = –500 mA; I_{Bon} = –0.5 mA; I_{Boff} = 0.5 mA	–	400	–	ns
t_{off}	turn-off time		–	1500	–	ns

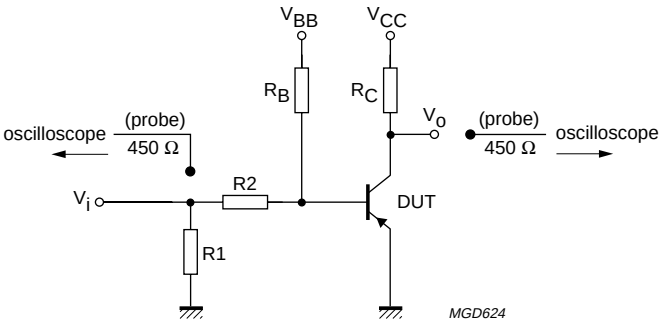
Note

1. Pulse test: $t_p \leq 300\ \mu s$; $\delta \leq 0.02$.



V_{CE} = −10 V.

Fig.2 DC current gain; typical values.

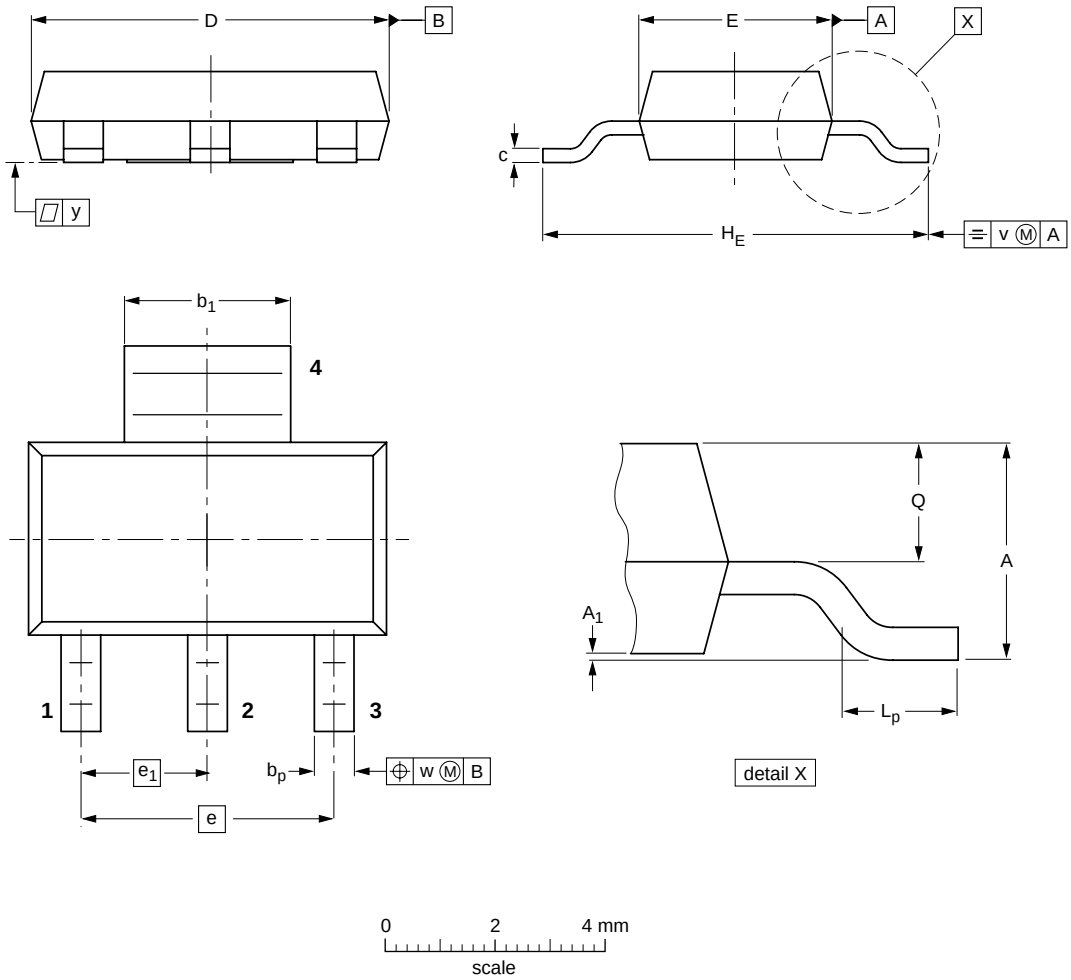


V_i = −10 V; T = 200 µs; t_p = 6 µs; $t_r = t_f \leq 3$ ns.
R1 = 56 Ω; R2 = 10 kΩ; R_B = 10 kΩ; R_C = 18 Ω.
V_{BB} = 1.8 V; V_{CC} = −10.7 V.
Oscilloscope: input impedance Z_i = 50 Ω.

Fig.3 Test circuit for switching times.

PACKAGE OUTLINE

Plastic surface mounted package; collector pad for good heat transfer; 4 leads **SOT223**



DIMENSIONS (mm are the original dimensions)

UNIT	A	A_1	b_p	b_1	c	D	E	e	e_1	H_E	L_p	Q	v	w	y
mm	1.8 1.5	0.10 0.01	0.80 0.60	3.1 2.9	0.32 0.22	6.7 6.3	3.7 3.3	4.6	2.3	7.3 6.7	1.1 0.7	0.95 0.85	0.2	0.1	0.1

OUTLINE VERSION	REFERENCES				EUROPEAN PROJECTION	ISSUE DATE
	IEC	JEDEC	EIAJ			
SOT223						~~96-11-11~~ 97-02-28

DEFINITIONS

Data Sheet Status	
Objective specification	This data sheet contains target or goal specifications for product development.
Preliminary specification	This data sheet contains preliminary data; supplementary data may be published later.
Product specification	This data sheet contains final product specifications.
Limiting values	
Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or at any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability.	
Application information	
Where application information is given, it is advisory and does not form part of the specification.	

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